

Computer Assignment 4

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Exercise 4:1

4:1.1

To find out which variables and how they affect the survival of a patient admitted to ICU, We conducted the following model selection process. First, we built an empty model, which contains only intercept.

$$\text{logit}(P(V_2 = 1|V_{else})) = \beta_0$$

Then, we built the full model

$$\text{logit}(P(V_2 = 1|V_{else})) = \beta_0 + \beta_1 \times V_1 + \beta_3 \times V_3 + \dots + \beta_{21} \times V_{21}$$

By applying stepwise selection using AIC and BIC as criteria separately, and experimenting with forward, backward, and both-direction procedures, we ultimately obtained two well-performing models: the mstep1_AIC model including variables Patient(v1), Age(v3), Cancer(v7), Blood pressure(v11), Type of Admission(v14), Blood pH(v17), Blood carbon dioxide(v18) and Consciousness level(v21) and the mstep1_BIC model containing variables Patient(v1), Age(v3), Cancer(v7), Type of Admission(v14) and Consciousness level(v21). Detailed statistics are shown in Table 1, Table 2.

Table 1: Coefficients of mstep1_AIC

Variable	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.8295916	1.6821579	-2.871	0.004091	**
v1	-0.0023492	0.0009249	-2.540	0.011088	*
v3	0.0408356	0.0140573	2.905	0.003673	**
v7	2.6482445	0.9179755	2.885	0.003916	**
v11	-0.0135185	0.0072602	-1.862	0.062603	.
v14	3.4735028	1.0269790	3.382	0.000719	***
v17	2.1378570	0.8954219	2.388	0.016961	*
v18	-2.3571916	1.0061247	-2.343	0.019137	*
v21	5.0020222	1.1050330	4.527	5.99e-06	***

Table 2: Coefficients of mstep1_BIC

Variable	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-6.2584933	1.3523800	-4.628	3.70e-06	***
v1	-0.0021704	0.0008872	-2.446	0.014436	*
v3	0.0351432	0.0130024	2.703	0.006875	**
v7	2.3555626	0.8743101	2.694	0.007056	**
v14	3.5138204	0.9723334	3.614	0.000302	***
v21	3.9577986	0.8910241	4.442	8.92e-06	***

The most important factors has following characteristics,

1. Factors appear in both models: Variables included in both AIC- and BIC-based models indicate their importance.
2. P-value is important: Important factors are supposed to be statistical significant.
3. The odds ratio is far from 1: An odds ratio different from 1 implies that a one-unit change in this variable has a considerable impact on survival chance.

Thus, Cancer(v7), Type of Admission(v14) and Consciousness level(v21) are considered as the most important factors in our chosen model.

Table 3: Odds ratio with confidence interval of mstep1_AIC

Factor name	Factor mark	Odds ratio	Confidence interval
Cancer	v7	14.1292	[2.4783, 103.2527]
Type of Admission	v14	32.2495	[5.5852, 334.5645]
Consciousness level	v21	148.7136	[22.0767, 1800.1846]

Interpretation: In this model selected by AIC, all three factors(Cancer (v7), Type of Admission (v14), and Consciousness level (v21)) demonstrate Odds Ratios significantly greater than 1, and their 95% Confidence Intervals do not contain 1. This indicates that, while controlling for the effects of other variables in the model, all three factors are statistically significantly associated with an increased odds of "not survival". The Consciousness level (v21) has the highest OR (148.71), suggesting it has the biggest impact on the outcome and is the strongest risk factor. Cancer (v7) and Type of Admission (v14) are also significant risk factors.

Table 4: Odds ratio with confidence interval of mstep1_BIC

Factor name	Factor mark	Odds ratio	Confidence interval
Cancer	v7	10.5441	[1.9429, 66.6798]
Type of Admission	v14	33.5763	[6.3312, 309.4947]
Consciousness level	v21	52.3420	[11.2088, 418.4505]

Interpretation: The interpretation of model selected by BIC is similar to that by AIC. All three factors are positively associated with the odds of "not survival". And among them, the Consciousness level (v21) has the strongest influence with OR being the highest of 52.34.

4:1.2

We select mstep1_AIC as the final model for addressing questions 4:1.2 and 4:1.3. H-L test result of this model is as following:

Table 5: The result of Hosmer-Lemeshow Test

Statistics	Values
X-squared	8.6047
degrees of freedom	8
p-value	0.3767

Interpretation: H_0 of the Hosmer and Lemeshow method is that the model fits the data well. Here we can see X-squared = 8.6047, $df = 8$, and the p-value of HL test is $0.3767 > 0.05$, meaning that we can't reject the null hypothesis. It suggests that our model is well-calibrated.

4:1.3

The confusion matrix at a cut-off of 0.5 based on our chosen model mstep1_AIC is

Table 6: Confusion Matrix of cut-off 0.5

Actual	Predicted	
	Not-survive(1)	Survive(0)
Not-survive(1)	21	19
Survive(0)	5	155

The confusion matrix at a cut-off of 0.3 based on our chosen model mstep1_AIC is

Table 7: Confusion Matrix of cut-off 0.3

Actual	Predicted	
	Not-survive(1)	Survive(0)
Not-survive(1)	27	13
Survive(0)	18	142

The confusion matrix at a cut-off of 0.7 based on our chosen model mstep1_AIC is

Table 8: Confusion Matrix of cut-off 0.7

Actual	Predicted	
	Not-survive(1)	Survive(0)
Not-survive(1)	14	26
Survive(0)	1	159

The corresponding accuracy, sensitivity and specificity at these three cut-offs are

Table 9: Accuracy, sensitivity and specificity at three cut-offs

Cut-off	Accuracy	Sensitivity	Specificity
0.3	0.845	0.675	0.8875
0.5	0.880	0.525	0.9688
0.7	0.865	0.350	0.9938

Description and explanation:

As the classification cut-off value is progressively increased from 0.3 to 0.5 to 0.7, the model becomes more conservative and shows more tendency to predict a sample as the negative class(0, "survive"). Specifically, this is reflected in the Confusion Matrix, where the value of element n_{12} (False Negatives, FN) increases from 13 to 19 and subsequently to 26, while the value of n_{22} (True Negatives, TN) progressively rises from 142 to 155 and finally reaches 159. As the cut-off increasing from 0.3 to 0.5 to 0.7, the corresponding changes in metrics are:

- Accuracy increase first and then decrease, peaking at 0.5. Since a low cut-off(0.3) is too aggressive and tends to predict 1 even though the true class is 0. Meanwhile a high cut-off(0.7) is overly conservative, predicting 0 when true class is 1. Cut-off 0.5 balances the counts of True Positives and True Negatives and therefore maximizing the overall correctness.
- Sensitivity is continuous decreasing. Since $sensitivity = \frac{true_positives}{total_actual_positives} = \frac{n_{11}}{n_{11}+n_{12}}$, with the denominator remains the same, n_{12} increases, n_{11} decreases, sensitivity decreases as well.
- Specificity is continuous increasing. $Specificity = \frac{TN}{TN+FN} = \frac{n_{22}}{n_{21}+n_{22}}$, similarly, with the dominator fixed, n_{22} increasing, specificity value increases naturally.

4:1.4

The ROC curves of full model and our two chosen models(mstep1_AIC and mstep1_BIC) are shown as fig. 1.

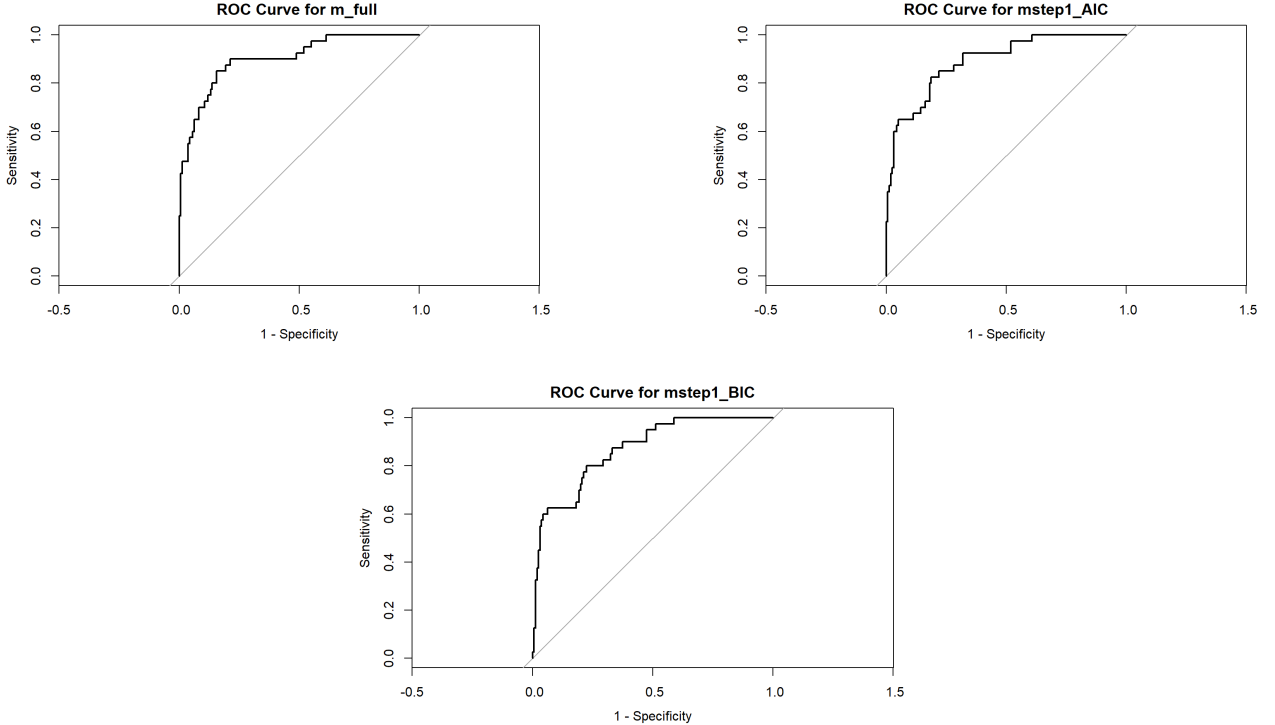


Figure 1: ROC curves of three models

The corresponding AUC values are:

Table 10: AUC values of three models

Model Name	AUC value
m_full	0.9012
mstep1_AIC	0.8920
mstep1_BIC	0.8677

We can see that the full model "m_full" performs best with the highest AUC of 0.9012.

4:1.5

After performing the Leave-One-Out-Cross-Validation method on the full model and our two chosen models(mstep1_AIC and mstep1_BIC), we got the LOOCV-adjusted AUC.

Table 11: LOOCV-adjusted AUC values of three models

Model Name	Original AUC	LOOCV-adjusted AUC	Absolute Difference
m_full	0.9012	0.7747	0.1265
mstep1_AIC	0.8920	0.8495	0.0425
mstep1_BIC	0.8677	0.8400	0.0277

Answer to the question:

The model with lowest AIC(mstep1_AIC) performs best based on LOOCV-adjusted AUC. Comparison:

- The m_full model achieves the highest initial AUC (0.9012). However, its AUC drops significantly to 0.7747 after the LOOCV procedure, losing 0.1265. This drop in performance indicates that the m_full model is overfit to the training data. It performs well on the data it was trained on but shows poor generalization ability when facing new data .
- The mstep1_AIC model has a slightly lower initial AUC (0.8920) than m_full. But its LOOCV-adjusted AUC (0.8495) is the highest among the three models, showing that the mstep1_AIC model maintains a high level of discrimination with the LOOCV procedure.
- The mstep1_BIC model demonstrates the smallest drop (0.0277) in AUC, showing that its performance is the most robust across different methods.

Exercise 4:2

4:2.1

We examined multiple decision trees using different **splitting methods (Gini and Information Gain)** and **complexity parameters (0.1, 0.07, 0.05, 0.03)** separately.

Gini Index

Measures impurity — the probability of misclassifying a randomly chosen element. Splits are chosen to maximize the reduction in Gini impurity. Computationally efficient and commonly used as the default in many algorithms.

Information Gain

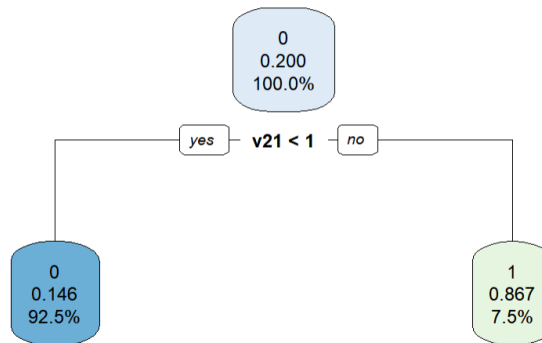
Measures how much a split reduces uncertainty (entropy). It selects the feature that provides

the greatest purity improvement using entropy-based calculations. Tends to favor features with many categories.

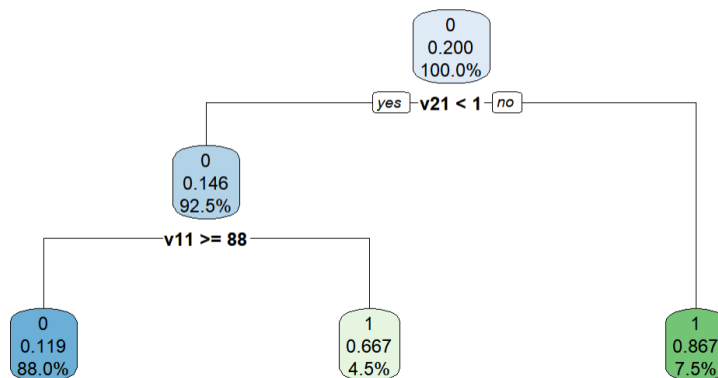
Complexity Parameter (cp)

A threshold that controls tree growth: a split is only made if it improves model purity by at least this value. Higher cp $\rightarrow$ simpler trees; lower cp $\rightarrow$ more complex trees.

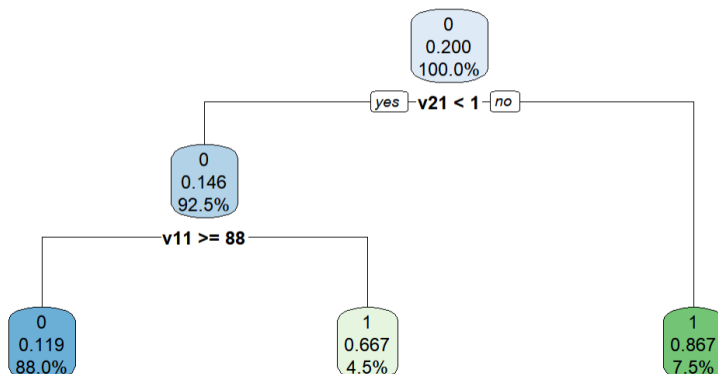
Decision Tree (split = 'information', cp = 0.1)



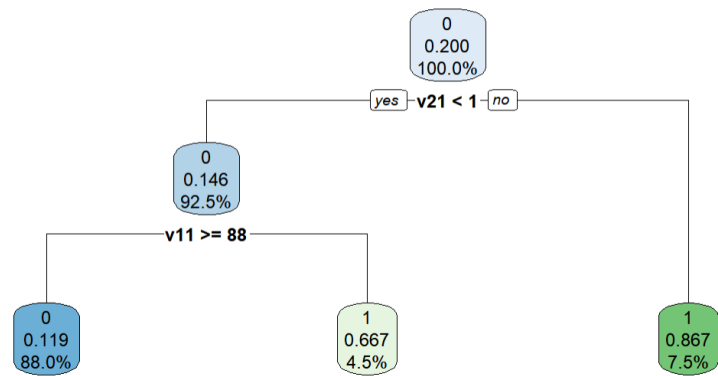
Decision Tree (split = 'information', cp = 0.07)



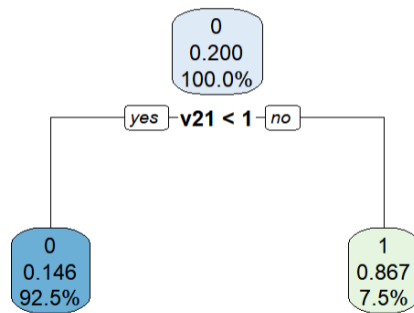
Decision Tree (split = 'information', cp = 0.05)



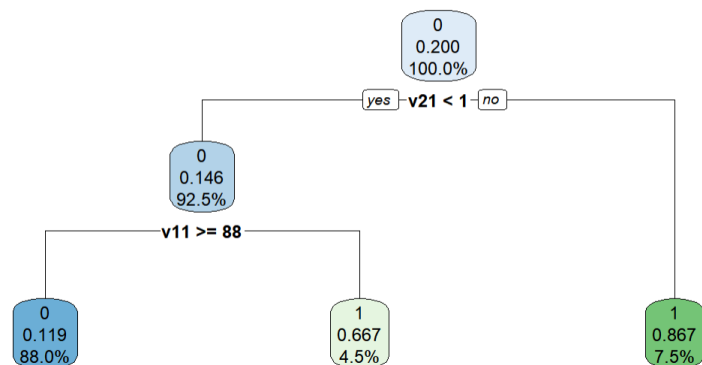
Decision Tree (split = 'information', cp = 0.03)



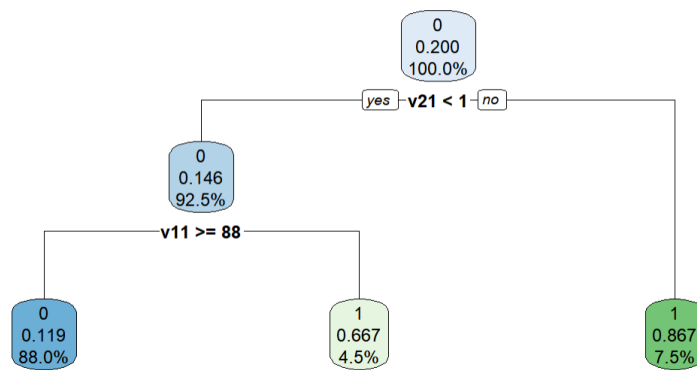
Decision Tree (split = 'gini', cp = 0.1)



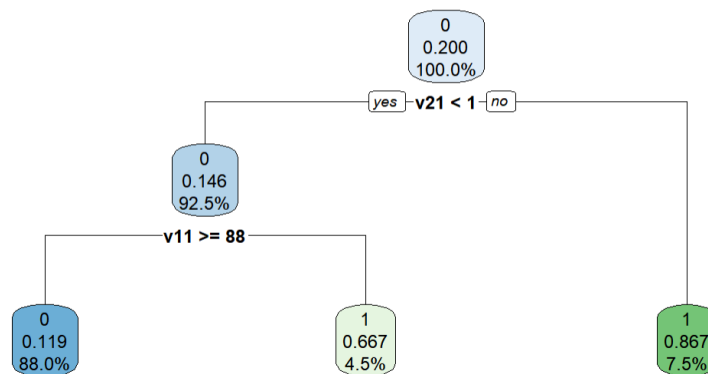
Decision Tree (split = 'gini', cp = 0.07)



Decision Tree (split = 'gini', cp = 0.05)



Decision Tree (split = 'gini', cp = 0.03)



Finally, we chose the **Decision Tree with split = "information" and cp = 0.07** as the "good" tree based on the interpretability and predictive power. Because two splits in the tree provide sufficient detail for predicting survival, and nodes are easy to interpret while capturing key patterns.

The description of the "good" decision tree (**Decision Tree with split = "information" and cp = 0.07**) is as follows:

Tree Structure & Splits:

1. Root Node: The tree first splits patients based on v_{21} (consciousness level).
 Left branch: $v_{21} = 0$ (awake patients) $\rightarrow$ 92.5% of cases
 Right branch: $v_{21} = 1$ (unconscious/coma patients) $\rightarrow$ 7.5% of cases
2. Second-Level Split:
 Only awake patients ($v_{21} = 0$) are further divided based on v_{11} (blood pressure):
 Left-left: $v_{11} \geq 88 \rightarrow$ 88.0% of awake patients
 Left-right: $v_{11} \leq 88 \rightarrow$ 4.5% of awake patients

Predictions & Probabilities:

1. Terminal Node 1 ($v_{21} = 0$ & $v_{11} \geq 88$): Predicted survival probability = 88.1% (mortality = 11.9%)
2. Terminal Node 2 ($v_{21} = 0$ & $v_{11} < 88$): Predicted survival probability = 33.3% (mortality = 66.7%)

= 66.7%)

3. Terminal Node 3 ($v_{21} = 1$): Predicted survival probability = 13.3% (mortality = 86.7%)

Key Insights:

1. Consciousness level (v_{21}) is the primary predictor: unconscious patients face very high mortality (86.7%).
2. Blood pressure (v_{11}) further stratifies awake patients: those with normal blood pressure (≥ 88 mmHg) have good survival (88.1%), while those with low blood pressure have poor survival (33.3%).
3. The tree with $cp = 0.07$ creates a clinically meaningful model: it's more informative than the simplest tree ($cp = 0.1$, which only used v_{21}) but avoids unnecessary complexity.

4:2.2

The AUC for the decision tree model (with split = "information" and $cp = 0.07$) is 0.724, and the ROC curve is shown in Figure 2.

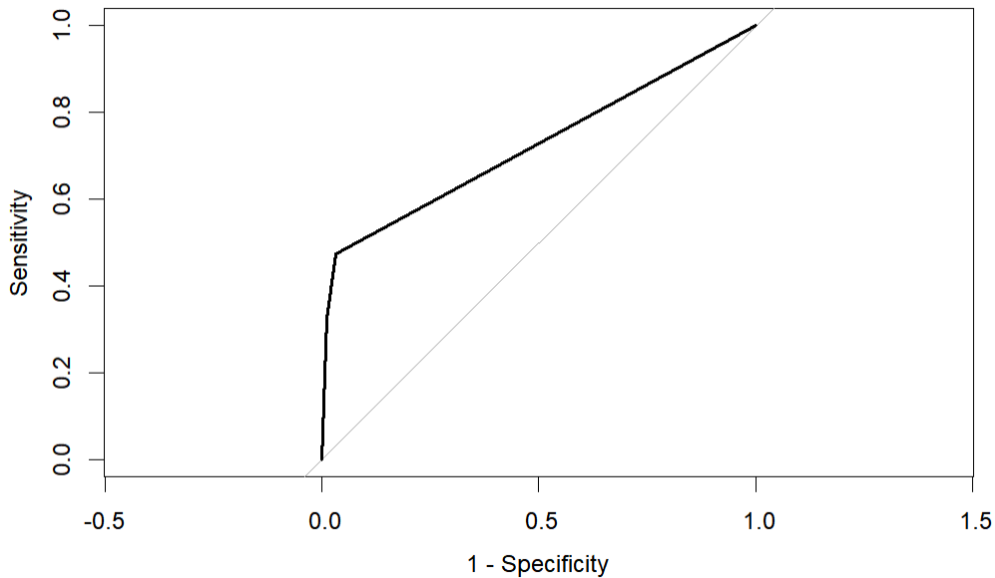


Figure 2: Decision Tree Model ROC Curve

Based on the provided AUC values, the multiple logistic regression model selected by AIC performs best with an AUC of 0.892. This is higher than both the BIC-selected logistic regression model (AUC = 0.868) and the decision tree model (AUC = 0.724). The ROC curves comparison is shown in Figure 4. Therefore, based on AUC as the measure of predictive power, the logistic regression model chosen via AIC is the top-performing model for predicting patient survival in the ICU study.

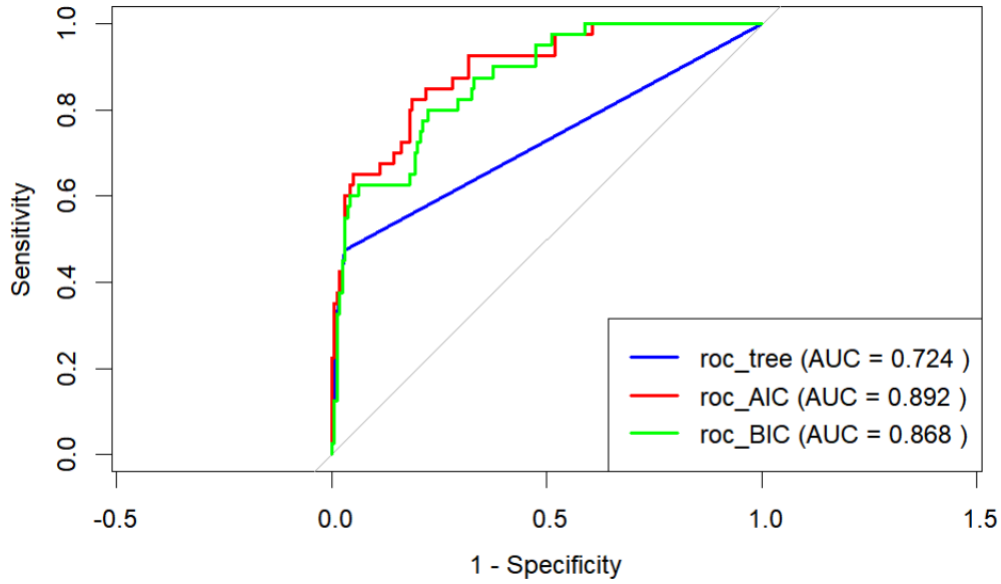


Figure 3: ROC Curves Comparison

4:2.3

Decision Tree Model: LOOCV-corrected AUC = 0.9186

Interpretation of the Result:

The decision tree model (using split = "information" and cp = 0.07) achieved a LOOCV-corrected AUC of 0.9186, indicating excellent ability to discriminate between surviving and non-surviving ICU patients. This strong performance, combined with the model's simplicity (using only consciousness level and blood pressure), suggests it could be clinically useful for risk stratification. The high cross-validated AUC also confirms that the model generalizes well and is not overfitting to the training data.

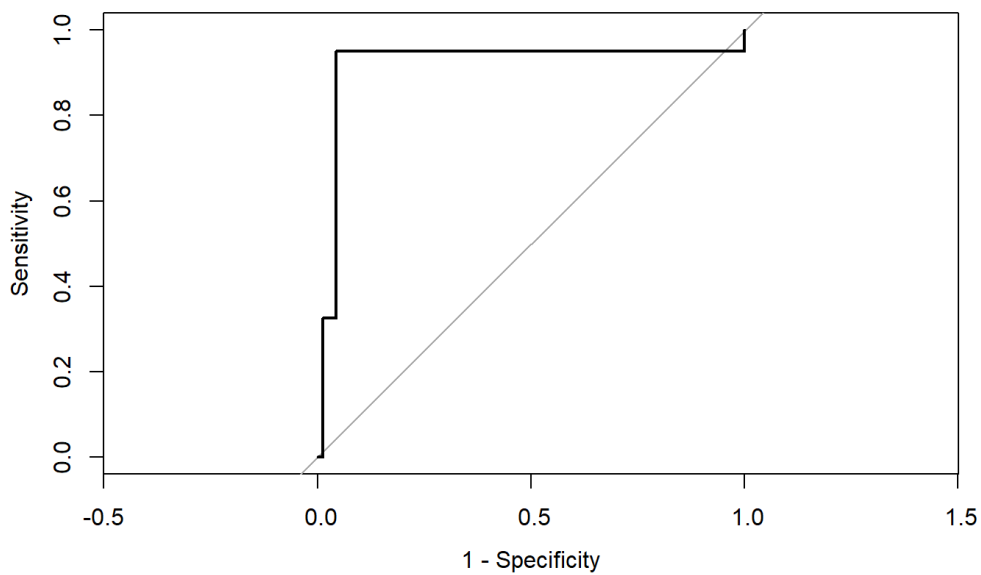


Figure 4: ROC Curves of LOOCV-adjusted Tree Model